Seeing the Map: Do Broadcast Minimaps Improve Live Win Probability in Professional League of Legends?

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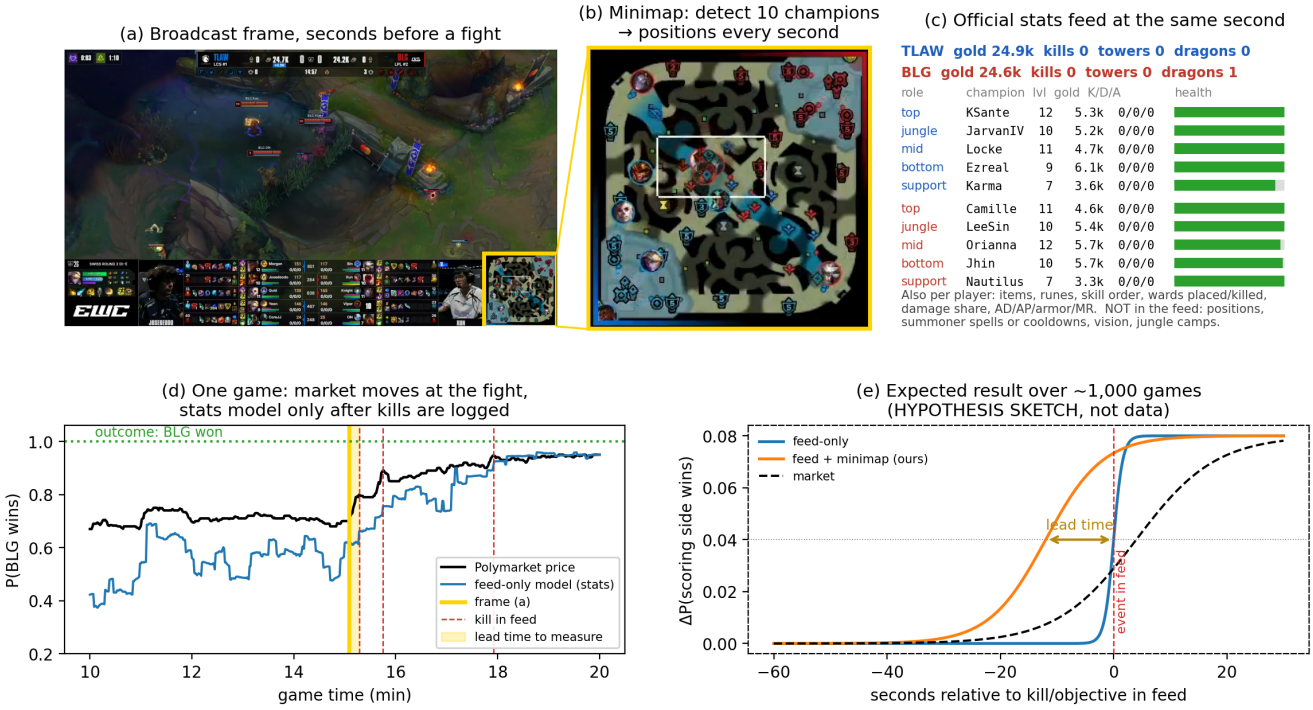


Figure 1: The idea on a real game (TLAW vs BLG, Worlds 2026; BLG won). (a) A broadcast frame a few seconds before the first teamfight. (b) The minimap, the only public source of player positions; we detect the ten champions in it every second. (c) What the official stats feed gives at that same second: team gold, kills, towers and dragons, and per player level, gold, K/D/A and health (plus items, runes, skill order, wards and combat stats), but no positions, summoner-spell cooldowns or vision. (d) Live win probability from our feed-only model (blue) and the Polymarket price (black). The fight is already visible on the minimap at the frame in (a) (yellow line); the market, whose traders watch the stream, moves then, while the stats-only model moves only once the kills reach the feed (red dashed lines). (e) A *hypothesis sketch, not data*, of the main result we will measure over ∼1,000 held-out games: the change in win probability of the scoring side, averaged over all kills and objectives aligned on the time they reach the feed. A feed-only model cannot move before $t = 0$; the lead time is how much earlier a model with minimap positions moves.

1 Problem and motivation

Given a professional League of Legends game up to time t, predict the probability that the blue team wins. We compare a model that sees only the official live-statistics feed with one that also sees the positions of all ten champions, extracted from the minimap in the broadcast video. Our hypotheses are that spatial features (1) reduce log-loss and Brier score, (2) matter most around contested objectives and in the mid game, and (3) move the prediction *before* the feed registers the outcome of a fight or objective.

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The minimap is the only public source of spatial state in professional games: we verified that the lolesports feed has no coordinates. Prior minimap work (DeepLeague [1]) detected champions but did not use positions for prediction, and its 56-class label set covers a 2017 champion pool. The project combines several course topics: convolutional detection, representation learning for spatial state, sequence models (GRU and transformer) over game time, and evaluation of calibrated probabilistic predictions.

2 Data

Sources (all verified). The lolesports esports API maps every game to its official VODs and, for 2026, its timestamp within the broadcast. The live-stats feed gives event-driven frames (median spacing 0.13 s) of team and player statistics and is retrievable back to at least 2023. Oracle's Elixir provides post-game outcomes as an independent check. Champion icons come from Riot's Data Dragon (173 champions).

Scale. In 2025, about 1,370 games from LCK, LEC, LCP, Worlds, MSI and First Stand have both a feed and an official VOD (LPL VODs are not in the API and are excluded); 2023–2025 together give roughly 4,000. At 2 frames per second this is about 5.8 M minimap frames for 2025, or about 90 GB of 190×190 crops. Videos are downloaded, cropped and deleted, so only crops are stored.

Labels. The feed names the ten champions and their sides, so at inference the detector only distinguishes ten known icons. We train it on synthetic minimaps made by pasting team-ringed Data Dragon icons onto real minimap backgrounds with distractors, and evaluate on about 300 hand-labelled real frames. Feed frames where a champion is dead give additional free negatives.

Alignment. On a pilot game, OCR of the broadcast clock aligned video to feed within about ± 0.5 s, which is negligible at a 1 Hz prediction rate.

3 Related work and relevance to the field

Sports analytics. In traditional sports, adding player-tracking data to box-score statistics changed how value is measured, for example expected possession value in basketball [4]. Esports has public tracking-like data only through the broadcast; we test whether the same step pays off.

Esports prediction. Live win prediction in Dota 2 [2, 3] and the win-probability graphics shown on League broadcasts rely on aggregate statistics. Whether spatial state adds information is an open question with direct use in broadcasting, coaching and betting markets.

Vision under weak supervision. The detector is trained without manual labels, from synthetic composites and the structured feed (the ten known champions, and death times as negatives). This is a reusable recipe for extracting state from any game broadcast that has a matching data feed.

4 Preliminary approach

4.1 Baselines

Feed-only model: logistic regression on gold difference, gradient-boosted trees on the full feed state, and a GRU over the feed history, the standard form of live win-probability models. **Betting market:** the live Polymarket price for the same game, reconstructed from public on-chain trades; it is an external baseline reflecting everything traders see, including the broadcast. A pre-game team-strength prior shows how much live data adds at all.

4.2 Method

1. **Feed-only baselines.** Gradient-boosted trees on the current feed state, and a GRU and a transformer over the feed history, trained on game-level splits with patch-aware held-out sets.
2. **Minimap detection.** A small YOLO-style detector on synthetic data, conditioned on the ten known champions, with tracking across frames.
3. **Spatial features.** Per-team centroid and spread, distance to the next objective, lane presence, and the number of champions near each contested area; and alternatively a learned encoder over a rasterised position map.
4. **Evaluation.** Log-loss, Brier score and reliability curves overall and per game phase; *lead time*, i.e. how many seconds earlier the spatial model's probability moves before feed events such as kills and objectives; and ablations by feature group.

4.3 What we measure

Quantity	Definition
Spatial gain $\Delta\mathcal{L}$	Log-loss (and Brier score) of the feed-only model minus that of the feed+minimap model on held-out games, with game-level bootstrap CIs; overall and in 5-minute game-time bins.
Calibration	Expected calibration error and reliability curves for both models [5].
Lead time τ	For each kill, objective or teamfight in the feed, the number of seconds before the event at which each model's probability first moves by more than a threshold in the eventual direction. The difference between models is the minimap's lead.
Event-conditional gain	$\Delta\mathcal{L}$ restricted to the 60 s before dragon, Baron and Herald contests, compared with quiet periods.
Detector quality	mAP and identity accuracy on 300 hand-labelled frames; position error in map units; fraction of frames with the minimap hidden.

The central tests are: is $\Delta\mathcal{L} > 0$ with a CI excluding zero; is it concentrated before contested objectives; and is τ positive, i.e. does the minimap anticipate what the feed only records afterwards.

4.4 What is new

- DeepLeague [1] detected champions on minimaps but never used positions for prediction; its labels cover 56 champions from 2017 and its weights are no longer available.
- Published live win-prediction models use statistics only [2, 3]. We give the first controlled measurement of what spatial state adds, aligned to the official feed to within about half a second.
- Lead time is a new evaluation axis for win-probability models: it asks not only whether a model is accurate but whether it knows earlier.
- We build a dataset of about 1,400 professional games per year with aligned feed and minimap positions, which does not currently exist publicly.

Timeline. Midterm (3 weeks): feed-only baselines on about 1,000 games, the synthetic-data detector with its accuracy on real frames, and the pilot alignment. Final (8 further weeks): full VOD processing, the combined model, the improvement over the feed-only baseline, lead-time analysis and ablations.

References

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